Matthew Sampson

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Accounts: LinkedIn Google Scholar Orcid GitHub

Education

Australian National University **Mt Stromlo, ACT**
Bachelor of Science Honours (Astrophysics) *2021 – Present*
Honours Thesis: "Numerically simulating cosmic ray propagation through turbulent molecular clouds."
Supervisory Team: Mark Krumholz, Roland Crocker, James Beattie

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Queensland University of Technology **Brisbane, QLD**
Bachelor of Science (Physics) with Distinction *2017 – 2020*
Cumulative GPA – **6.9/7**

Queensland University of Technology **Brisbane, QLD**
Bachelor of Mathematics (Applied and Computational) with Distinction *2017 – 2020*
Cumulative GPA – **7/7**

Publications/Conference Talks

Peer-Reviewed Papers: Stevenson, Simon, Sampson, Matthew et al. "The impact of pair-instability mass loss on the binary black hole mass distribution." *The Astrophysical Journal 882.2* (2019): 121. https://iopscience.iop.org/article/10.3847/1538-4357/ab3981
Conference Presentation: 2020 "Rapid and Effective Population Calibration", Australian National University student conference.
Conference Presentation: 2018 "Pair-instability supernova and the binary black hole mass distribution", Australian National Institute for Theoretical Astrophysics conference, Melbourne.

Research Experience

2020 - RESEARCH ASSISTANT QUT (MATHEMATICAL MODELLING) **Brisbane, QLD**
Working as a research assistant at the Queensland University of Technology working with Professor Kevin Burrage and Dr Brodie Lawson on calibration of models involving techniques from the field of information geometry.

2019 – 2020 - ICRAR SUMMER SCHOLAR **Crawley, WA**
Awarded a fully funded scholarship at the International Center for Radio Astronomy and Research (ICRAR) for the summer of 2019/20. Working under the supervision of Professors Chris Power and Peter Quinn investigating the structure of dark matter haloes through the use of observational data, and SPH simulations.

2019 – 2020 - RESEARCH ASSISTANT QUT (DATA ANALYSIS/ECOLOGY) **Brisbane, QLD**
Employed as a research assistant at the Queensland University of Technology working under the supervision of Associate Professor Jennifer Firn. Tasks involve using local HPC infrastructure to run statistical models using R on ecological data

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2018 – 2019 - ASTROPHYSICS SUMMER SCHOLAR **Hawthorne, Vic**
Awarded a scholarship in the summer of 2018/19 from the Swinburne Centre for Astrophysics and Supercomputing to complete a funded research project. Worked under the supervision of Dr Simon Stevenson and Dr Jade Powell to study the effect of Pulsational Pair Instability supernova on the binary black hole merger rate. This was done using the HPC facility OzStar and the population synthesis code COMPAS. https://compas.science/

2018 - RESEARCH ASSISTANT QUT (MEDICAL PHYSICS) **Brisbane, QLD**
Formerly employed as a research assistant at the Queensland University of Technology working under the supervision of Professor Soniya Yambem. Involved in the production and testing of organic biosensors used to detect trace amounts glucose from external sources such as saliva.

Teaching Experience

Experience teaching the following units at QUT where my role was the delivery of content, marking of assignments in an equitable manner and engaging students with new content such as programming languages and higher level statistics for the first time.

- SEB104 Grand Challenges in Science (3 Semesters)
- SEB113 Quantatative Analysis (3 Semesters)
- SEB115 Experimental Science (2 Semesters)
- PVB101 Physics of the Very Large (2 Semesters)
- BVB204 Ecology (2 Semesters)

Technical Skills

Computational and mathematical skills include;

- **LaTeX** – $[Highly\ Skilled]$
- **Python** – $[Skilled]$
- **C++** – $[Skilled]$
- **MATLAB** – $[Skilled]$
- **R** – $[Skilled]$
- **Linux OS** – $[Skilled]$
- **Git** – $[Basic\ Skills]$
- **Numerical Analysis**
- **Linear Algebra**
- **Computational Fluid Dynamics**
- **Differential Equations**
- **Mathematical Analysis**
- **Multivariable Calculus**

Scholarships and Awards

- 2021 Awarded the Bok Honour Scholarship in Astrophysics at the RSAA $7000
- 2020 Awarded equal first place prize for best science talk at ANU RSAA student conference
- 2020 Awarded best presentation PVB302 Quantum Physics poster conference $50
- 2019 Awarded a place into the Deans Scholars Program QUT
- 2019 ICRAR-Pawsey Centre Vacation Scholar $10000
- 2019 Deans List of Commendation for academic excellence
- 2018 Swinburne Centre for Astrophysics and Supercomputing Summer Vacation Scholarship $4000
- 2018 Deans List of Commendation for academic excellence
- 2017 Accepted into QUT's College of Excellence for high acheiving students
- 2017 Deans List of Commendation for academic excellence

Extracurricular Activities / Interests

- President of the QUT Physics Society
- Volunteered for the student run tutoring support network for STEM students STIMulate
- Physics representative for QUT's 'Science students as partners' program
- Avid retro computer game fan, currently working on a text based adventure game using C++

Referees

- Provided as per request